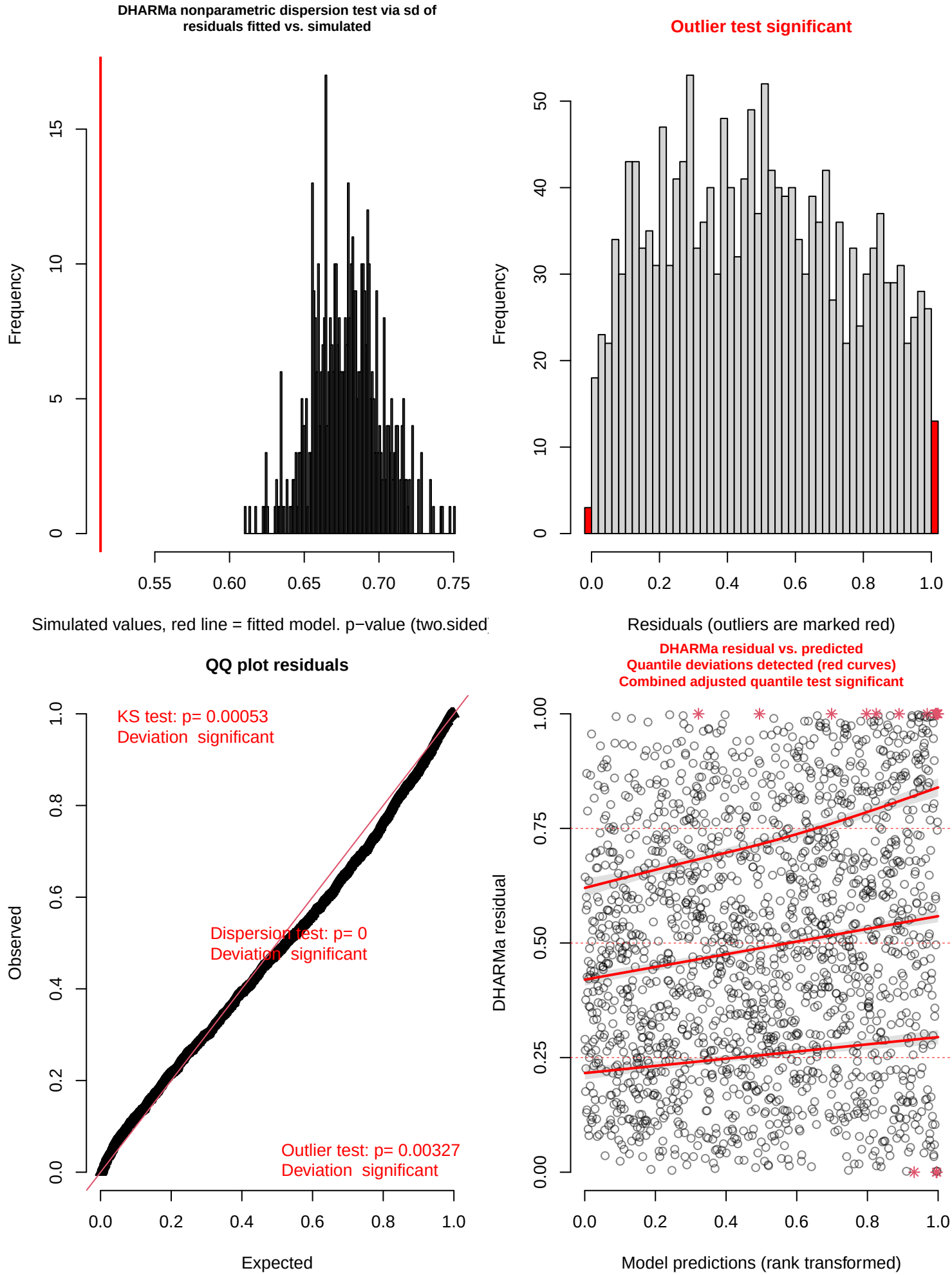
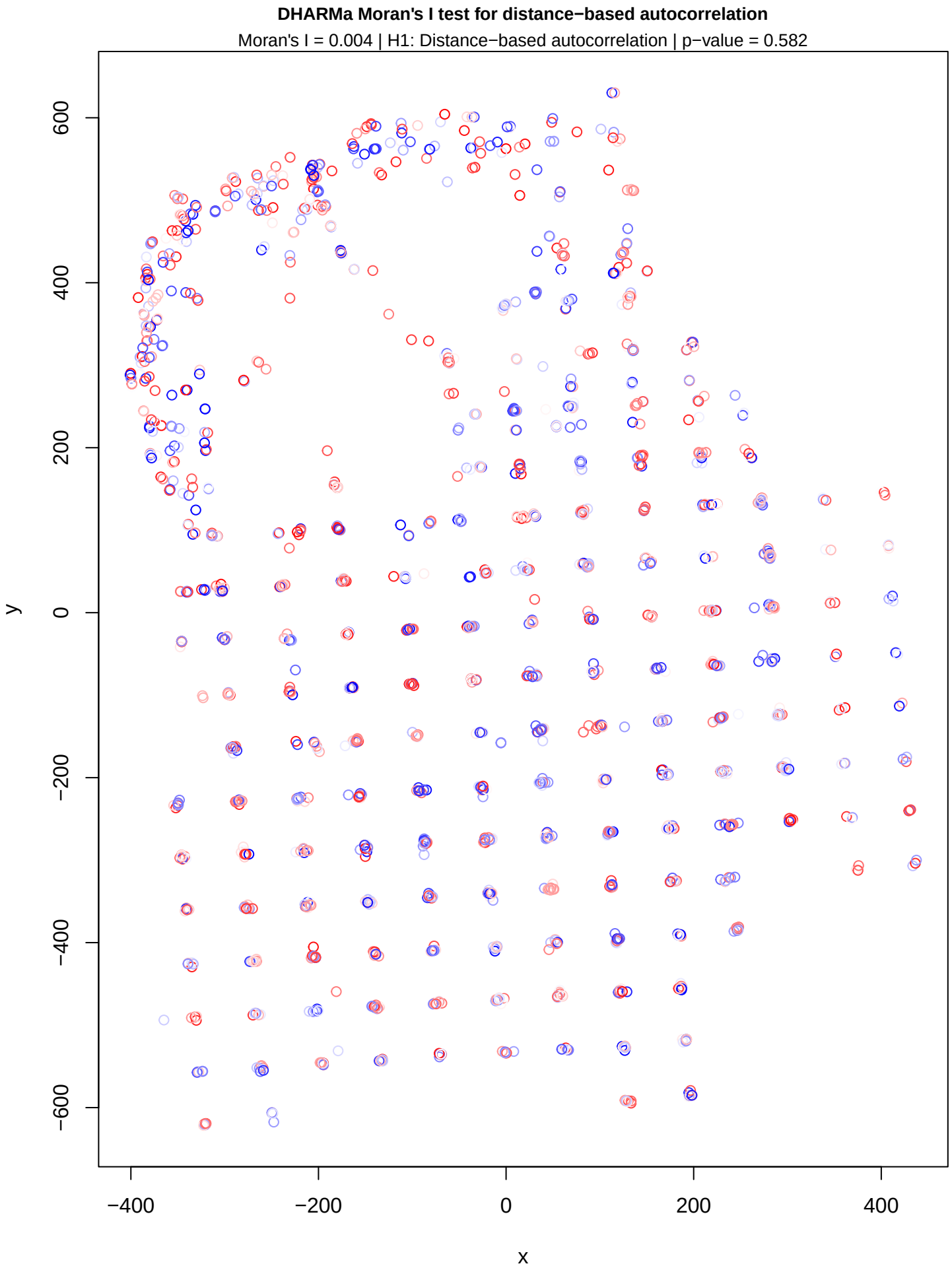
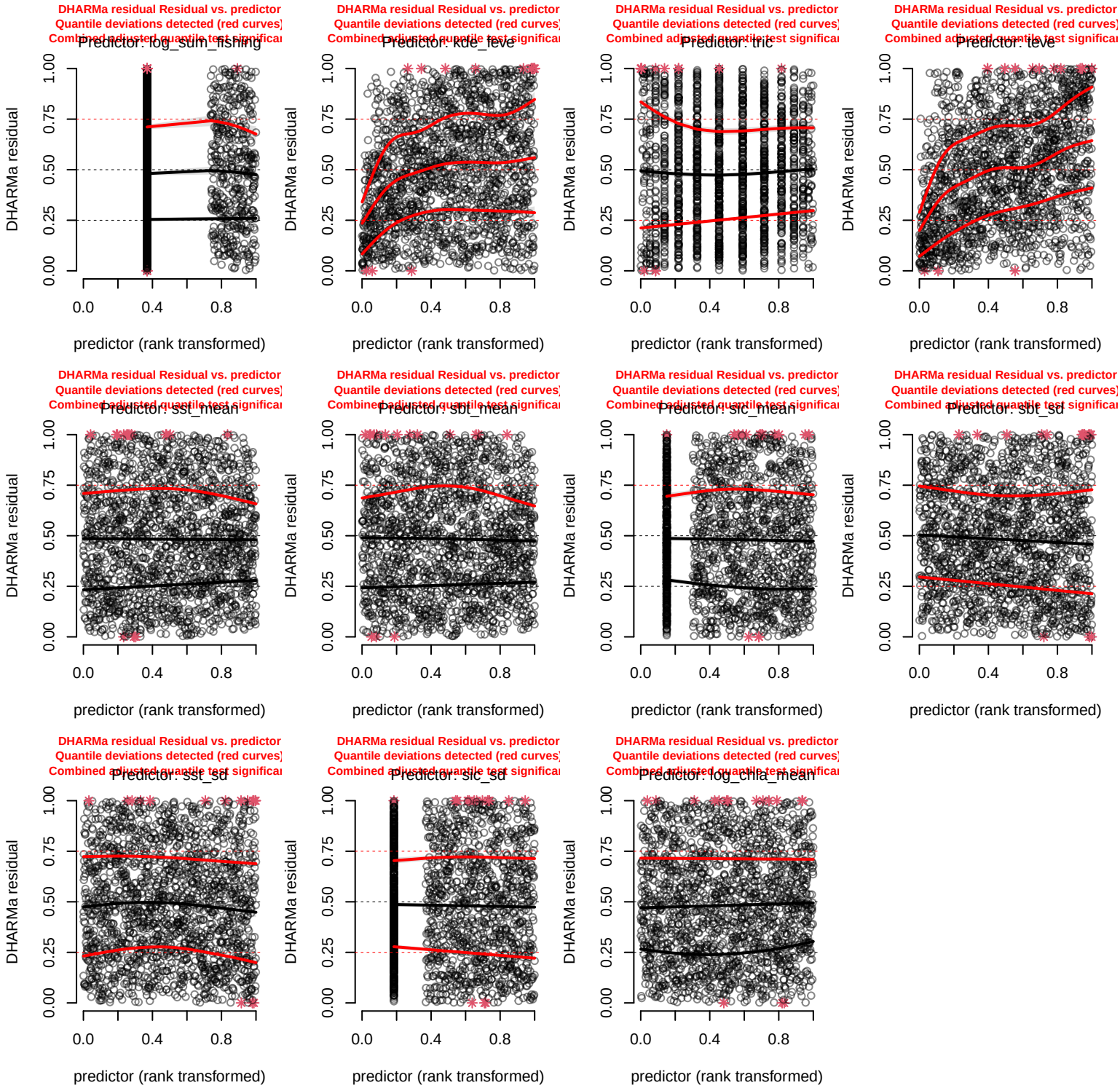
```
--------------------------------------------------------------------
MODEL SELECTION AND VALIDATION REPORT ( fishing_kde_fric model )
--------------------------------------------------------------------

## OBSERVATIONS:1755

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( fishing_kde_fric model )

Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

Conditional model:
                coef.est coef.se
(Intercept)        −0.10    0.13
log_sum_fishing    −0.03    0.03
sbt_mean            0.10    0.07
sbt_sd              0.04    0.04
sst_sd              0.08    0.03
sic_mean           −0.04    0.06
log_chla_mean       0.00    0.03
depth              −0.21    0.04

Dispersion parameter: 0.78
Spatiotemporal AR1 correlation (rho): 0.90
Matérn range: 216.92
Spatiotemporal marginal AR1 SD: 0.64
REML criterion at convergence: 2262.341

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                  df      AIC  delta_AIC
spatiotemporal_random_fields_ar1                  12 4548.681   1.894040
spatiotemporal_random_fields_ar1_with_anisotropy 14 4550.575  49.922809
spatial_random_fields                             11 4600.498   2.331148
spatial_random_fields_with_anisotropy             13 4602.829  82.158948
spatiotemporal_random_fields_iid_with_anisotropy 13 4684.988   1.360402
spatiotemporal_random_fields_iid                  11 4686.349 147.790103
error_only                                         9 4834.139         NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.00053
Deviation significant
Dispersion test: p= 0
Deviation significant
Outlier test: p= 0.00327
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.004 | H1: Distance−based autocorrelation | p−value = 0.582
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: log_sum_fishing
Predictor: kde_leve
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8